

A Sensorless Markerless Screening Framework for Identifying Kinematic Patterns Associated with **Elevated Injury Risk** Across Multiple Athletic Exercises

Shubham Shirodkar

MSc Artificial Intelligence and Data Science (COM726)

Southampton Solent University · Supervisor: Dr Raza Hasan

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The Screening Gap

Injury screening today lives at two extremes — neither suited to real-world coaching and rehabilitation environments.

01

Lab-Based Motion Capture

Vicon, Qualisys and marker-based systems provide accuracy — but at £100k+, they are inaccessible to most training environments and require specialist operators.

02

Consumer Wearables

Phone apps and fitness devices are accessible and inexpensive, but are generally unvalidated for kinematic screening decisions.

03

The Unmet Need

Coaches, physiotherapists, and athletes need something accessible AND trustworthy — validated enough to inform screening decisions.

This dissertation addresses the gap between the two extremes.

Screening, **NOT** Prediction

WHAT THIS WORK IS

Detects kinematic patterns previously associated with injury risk in the biomechanics literature.

Operates on personalised deviations from cohort-informed baselines.

Provides transparent, auditable screening flags with counterfactual explanations.

WHAT THIS WORK IS NOT

A clinical prediction system or injury forecasting tool.

A diagnostic classifier producing individual risk scores.

A replacement for clinician judgement — clinical interpretation remains with qualified practitioners.

This distinction determined every methodological choice.

Datasets and Cohorts

All publicly available secondary research datasets — ethics-clean, fully reproducible.

Dataset	Role	Cohort Size
REHAB24-6	Primary screening cohorts (squat + lunge)	9 squat subjects (98 reps) 7 lunge subjects (61 reps)
OpenCap (Uhlrich et al. 2023)	Ground-truth validation cohort — the trust anchor	8 subjects 48 drop-jump trials
Penn Action (Zhang et al. 2013)	Cross-cohort exploratory squat data	10 subjects (single-rep)

Design rationale: Public datasets provide ethical clearance for secondary use, avoid recruitment bias, and ensure full reproducibility. This is a validation-and-framework study rather than a clinical trial — I own the small n openly, and the design is intentional.

End-to-End Framework

INSERT FIGURE HERE

Figure 2.1 — Pipeline Architecture Diagram

*Replace this placeholder with your pipeline diagram
(Video → MediaPipe Pose → Biomarker Extraction → Screening Layer + XAI,
with Ground-Truth Validation branch)*

Figure 2.1 — End-to-end sensorless kinematic screening pipeline

Consistent Discrimination Across Exercises

Both exercises show large, statistically robust effect sizes on peak knee flexion.

Figure 4.2
Squat Forest Plot

SQUAT
Cohen's d = 1.73

Figure 5.2
Lunge Forest Plot

LUNGE
Cohen's d = 1.69

Both effect sizes are large with confidence intervals clear of zero. Cluster-aware bootstrap (subject-level) mitigates within-subject inflation.

Ground-Truth Validation

Bland-Altman comparison against marker-based motion capture (OpenCap cohort).

Figure 6.1
Bland-Altman Biomarker Panels

PEAK FLEXION BIAS

+10.52°

95% LIMITS OF AGREEMENT

[-5.54°, +26.58°]

CHARACTER

Systematic

Predictable

Correctable

This characterisation is the foundation of the uncertainty framework. Without it, no downstream screening decision would be defensible.

Uncertainty-Weighted Screening

Inverse-variance weights derived directly from validation error.

PEAK FLEXION

57.15%

CONTACT / START FLEXION

22.63%

RANGE OF MOTION

15.30%

VELOCITY

4.92%

Cross-exercise sensitivity: less than 8.6% variation — weights transfer across squat and lunge.

Counterfactual Explainable AI

Faithful-by-construction explanations for rule-based screening decisions.

XAI Worked Example Figure

Worked example: Squat repetition 6, subject PM_113

DEPTH MARGIN

17.77° *deviation from personal baseline*

ROM MARGIN

7.83° *deviation from personal baseline*

VELOCITY MARGIN

19.82°/frame *deviation from personal baseline*

The explanation IS the decision — not a post-hoc approximation of a black-box.

Pre-Registered Null Result

Temporal deep learning did not outperform peak-flexion baselines — an honest, informative finding.

Model	Squat Balanced Accuracy	Lunge Balanced Accuracy
Peak-flexion baseline	81.36%	81.50%
LSTM (Scheme A)	53.79%	53.78%
LSTM (Scheme B)	50.00%	39.17%

Why this strengthens — not weakens — the framework.

This was a pre-registered null result. In low-data regimes, a strong peak-flexion baseline dominates sequential deep learning models. This finding directly justifies the 57.15% peak-flexion weight in the uncertainty framework — the LSTM null tells us where the signal genuinely lives.

Contributions and Boundaries

FOUR NOVEL CONTRIBUTIONS

01 Cross-Exercise Screening Pipeline

Sensorless kinematic screening across squat, lunge, and drop-jump.

02 Failure-Mode Taxonomy

Real errors documented and classified — not hypothetical failure modes.

03 Uncertainty-Weighted Transfer

Screening weights generalise across exercises (< 8.6% variation).

04 Faithful-by-Construction XAI

Counterfactual explanations that ARE the decision — not post-hoc approximations.

HONEST BOUNDARIES

► Screening, not prediction

No prospective outcome data — no injury forecasting claim.

► Secondary datasets

Public data; not a self-collected prospective clinical trial.

► Track A foundation

Temporal DL and digital twin work remain future scope.

► Modest cohorts

n = 9, 7, 8, 10 — validation-and-framework, not generalisability study.



Thank you.

Questions?

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